

AMBER SHI

RESUME - MUSIC, SCIENCE & TECHNOLOGY
OCTOBER 2026

MUSIC TECHNOLOGY & COMPOSITION PROJECTS

Hearing California (2025-Present)

- Composed and produced an original piece in Ableton Live, layering MIDI instruments and effects to build the arrangement
- Used AI image generation to create visuals of California landscapes and animals as source material for the video
- Programmed 3D visuals in Max/Jitter that respond to the music, syncing the video's movement to the song's tempo

Signal, Movement, Resonance (2026)

- Designed an interactive system in Max/MSP that turns human movement into synthesized sound, played back through physical resonators
- Used frame differencing and MediaPipe pose tracking to map the position and intensity of left-wrist movement to melodic and percussive sounds in real time
- Explored how synthesized audio changes when it resonates through acoustic objects rather than speakers
- Built the resonators from recycled materials (cardboard boxes, a $5 secondhand guitar body, and a tambourine), each driven by a speaker and power amplifier

MUSIC EDUCATION & TRAINING

Private Piano Study - The Calder School (2014 - Present)

- Instructor: Ludwig Calder

- Trained for 12 years in performance technique, sight-reading, aural skills, and music theory
- ABRSM: **Grade 8 Piano (Distinction)** (2025)
- MTAC Certificate of Merit: **Advanced Level** (2023)
- United States Open Music Competition (USOMC)
  - **1st place** Open Solo (2020), performing Kabalevsky, Slow Waltz, Op. 39 No. 23
- Selected Repertoire:
  - Beethoven, Piano Sonata No. 14 in C-sharp minor, Op. 27 No. 2 (Moonlight), III. Presto agitato
  - Beethoven, Piano Sonata No. 8 in C minor, Op. 13 (Pathétique), I. Grave - Allegro di molto e con brio
  - Chopin, Étude in C minor, Op. 10 No. 12 (Revolutionary)
  - Chopin, Étude in G-flat major, Op. 10 No. 5 (Black Key)

Private Music Technology Study (2024 - Present)

- Instructors: Houyu Chen, Annie Gao

- Compose, arrange, and edit original music in Ableton Live
- Program interactive audio patches in Max/MSP

Georgia Tech - Music Technology Pre-College Program, School of Music (Summer 2026)

- Programmed a custom drum sequencer in VCV Rack
- Built an interactive Arduino controller to adjust scale, filter, delay, reverb, and sequencer tempo in real time

Northern Arizona University - Flagstaff Piano Festival (June 2025)

- Instructors: Silvan Negrutiu, Aimee Fincher, Kenneth Osowski

- Performed Beethoven's Moonlight Sonata, 3rd mvt. (Presto agitato), in a masterclass with Sofya Gulyak, winner of the Leeds International Piano Competition
- Refined technique for rapid arpeggios and expressive phrasing

National Student Leadership Conference (NSLC) - Music Industry Program, held at UCLA (Summer 2024)

- Recorded and mixed audio in Ableton Live and other DAWs
- Co-wrote, recorded, and produced an original single with a team of four

MUSIC TECHNOLOGY & COMPOSITION PROJECTS (CONT.)

Mood Swinger (2026)

- Built an interactive 3D brain model inspired by the two brain hemispheres, where the left side controls rhythm and the right side controls pitch
- Programmed an ESP32 microcontroller (Arduino IDE) to read capacitive touch and knobs, sending serial data to Max/MSP
  - Knobs control mood, tempo, and filter cutoff
  - LED light that changes color and blinks with the tempo
- Designed three moods (happy, angry, and confused), each with its own wave form, melody, and drum sounds

Overtone Dialogue (2025)

- Composed an original piano piece and recorded a live performance of it
- Synthesized guitar, bass, chime, and pad sounds in Max/MSP
- Programmed Max/MSP to analyze the high-frequency content of the live piano and use it to control the pitch and volume of the synthesized instruments

MUSIC PSYCHOLOGY PROJECT

Emotion Builder (2026-Present)

- Inspired by *Music, Thought, and Feeling*, designed and released an iOS app where users log their mood, listen to three randomly selected songs, and rate how they feel afterward
- Built the app with an AI coding assistant (Claude Code); led the concept and UI, testing, and App Store release
- Analyzed 40+ journal entries to study how musical tempo and key relate to emotion
- Found that 63% of users reported improved mood after major-key songs

SELECTED PERFORMANCE EXPERIENCE

Carnegie Hall - Symphonic Series for Bands and Orchestras, New York, NY (June 2026)

- Performed cymbals and snare drum in Bizet's Carmen Suite
- Performed a four-hand piano arrangement of Dvořák's New World Symphony

Bay Area Youth Music Society (2023 - Present)

- Organize choir repertoire and lead monthly rehearsals
- Perform piano and vocals at senior living communities twice a month
- Performed primo part of Dvořák's Slavonic Dances, Op. 46 No. 5 (four hands)

VOLUNTEER SERVICE AWARDS

Senior Living Performances at Bay Area Youth Music Society

- American Volunteer Service Award **Group Gold** (2026)
- National Service Honors **Group Gold** (2025)
- President's Volunteer Service Award **Bronze** (2024)

OTHER PERFORMING ARTS

Dance Arts (2016 - Present)

- Valley Christian High School Vivid Dance Team (2024 - Present)

- Instructors: Michelle McLaughlin, Colette Evans, Nigel Shum

- West Valley Dance Company Competition Team (2021 - 2024)

- West Valley Dance Company Performance Team (2016 - 2021)

- Trained in hip hop, jazz, contemporary, ballet, and lyrical, with a focus on musicality and the connection between music and movement
- Edited and mixed music for a hip hop solo (2026 - 27 season)
- National awards include:
  - **2nd place** Varsity Large Character - Contest of Champions National Dance Competition (2026)
  - **1st place** Teen Large Hip Hop (Finals) - Energy National Dance Competition (2024)
  - **1st place** Teen Large Jazz Line - Energy National Dance Competition (2023)

Choir - Soprano/Alto (2016 - 2023)

- Instructors: Ms. Aeja (Action Day Schools); Ivan Utomo (Valley Christian Schools)

- Sang soprano and alto in school choirs over seven years
- Performed in school assemblies and seasonal concerts

MUSIC COURSEWORK

AP Music Theory (2025): 5